

SENG1050 Assessment 3

Web-Based Assignment 2 (25%)

Milestone 1: Checked starting from Week 8 Labs

Milestone 2: Checked starting from Week 10 Labs

Due Date: 11:59 pm on 24 Oct (Week 12)

[Outline](#) - [General Requirements](#) - [Web Server Setup and Website Import](#) - [Genre Pages](#) - [Data Collection Page](#)
- [Responsive Web Design](#) - [Form Submission Script](#) - [Coding Style](#) - [Submission](#) - [Marking Guide](#)

The objectives of this assignment are as follows:

1. To exercise your ability to create web pages using HTML and CSS.
 2. To transform the XML documents from Assessment 2 (Web-Based Assignment 1) into HTML using PHP.
 3. To test your skill in deploying a website in a self-hosted server environment.
 4. To evaluate your expertise with JavaScript for client side processing.
-

1. Outline

Following the success of the initial website launch, the *FeverGamez* board have decided to expand the website to facilitate selling second-hand products submitted by customers from all over the world. More precisely, your project is now to extend your website to allow people to complete a form listing their second-hand products with all the details required by *FeverGamez*. Several website improvements should also be included.

Your website should still be simple, friendly and attractive. This can be achieved by satisfying the requirements and completing the tasks in the following sections:

- [Section 2: General Requirements](#)
 - [Section 3: Web Server Setup and Website Import](#)
 - [Section 4: Category Pages](#)
 - [Section 5: Data Collection Page](#)
 - [Section 6: Responsive Web Design](#)
-

2. General Requirements

2.1 Web Server Requirements

Your website must run on an *Apache* web server which uses *PHP* to implement the server-side behaviour. This website must be accessible from the URL: *http://fevergamez.com*.

Your website must maintain a proper directory. All files must be stored in a directory named *fevergamez*. For clarity, all images included in the website must be stored in the *fevergamez/data/images* directory; all xml files included in the website must be stored in the *fevergamez/data/xml* directory; and all template files included in the website must be stored in the *fevergamez/themes* directory. You may include additional directories according to your discretion.

2.2 HTML Requirements

All the HTML pages (**generated using PHP**) that form your website should satisfy the following:

- Coded according to the HTML Living Standard.
- Contain the full HTML document structure, i.e., include `<!DOCTYPE>`, `<html>`, `<head>`, `<meta>`, `<title>` and `<body>`.
- There should be complete navigation capabilities across your website, such that:
 - Each webpage should contain hyperlinks to every other webpage.
 - The hyperlinks are implemented using *relative* URLs.
- The footer of each page should contain a copyright statement including your name, student number, email address, and degree.

2.3 CSS Requirements

In general, in order to draw your target audience, make sure that your website considers factors such as navigability, content layout, element positioning on screen, text size, colour contrast, readability, consistency, and accessibility. All visual formatting of the Web site will be implemented using Cascading Style Sheets (CSS). You will be using external style file, document-level style sheet, and inline styles in this assignment.

Create at least one external style file named `style.css`. This external style file should achieve the following CSS requirements to all your web pages:

- Left and right side margin of at least **30px** for the main content (body) of all webpages. **Note that this is improved from the previous assignment to cater for responsive design.**
- A nice background for the webpages. The text must be readable with the background.
- The text colours of a link before they are visited, after they are visited, and during a user action must be different from each other. The link must **be italicized** when you move the cursor over them. **Note that this is improved from the previous assignment to cater for responsive design.**
- Include at least one *id* and one *class* selector.
- Hooked to at least one `<div>`, used to provide some interesting styling (your choice of borders, background, floating alignment, etc.)
- Hooked to at least one `<span>`, used to provide some formatting (e.g., a colour or font change) for a few words of text.

You must also include at least one document-level style sheet and at least one inline style to the Homepage (`index.php`). Note that document-level style sheet is used to apply styles that are unique to the particular page, whereas inline styles should only be used for one-off visual effects.

3. Web Server Setup and Website Import

3.1 Web Server Setup

This subsection involves setting up the Apache web server with XAMPP. This could be easily achieved by following the steps outlined in Task 2 of Lab 6.

3.2 Website Import

Upon completing [subsection 3.1](#), you will import your website from Assignment 1. This can be achieved through the following steps:

- Create the `fevergamez/data/images` directory. Move all images in the website to this directory. Remember to change the image `src` attributes in your HTML code to point to the new location.

- Create the *fevergamez/data/xml* directory. Move all xml files in the website (`xmltemplate.xml`, `xmldata1.xml`, and `xmldata2.xml`) to this directory.
- Create the *fevergamez/theme* directory. Copy your external stylesheet *style.css* into this folder.
- In the *fevergamez/theme* directory, create a PHP template file called `template.php`. In this file, create a template that links to the external stylesheet and data of your HTML pages. *[Hint: refer to Lab 7 Task 1.]*
- Convert all HTML pages from the previous assignment (`index.html`, `about.html`, and `contact.html`) into PHP files (`index.php`, `about.php`, and `contact.php`) such that they will be generated using the PHP template (`template.php`). *[Hint: refer to Lab 7 Task 1.]*

It is recommended to double check that your webpages satisfy the requirements in [subsection 2.2](#) and [subsection 2.3](#).

Milestone 1

Upon completing [section 3](#), approach your lab instructor to check your first milestone during the labs between Week 8 and Week 12. The first milestone aims to check that you have completed your setup of the Apache web server and that you have Successfully import your website from Assignment 1.

The following will be checked:

- The directory structure of your virtual host should be as specified in [subsection 2.1](#).
- The website is accessible from *http://fevergamez.com*.
- The default webpage for your website should be your homepage from Assignment 1.
- All webpages in your website should be loading all linked resources (hyperlinks, images, etc.) correctly.
- Your external CSS should be correctly linked.

Fulfilling the criteria listed above will grant you 1% of your Assessment 3 score.

4. Genre Pages

4.1 Renaming XML Data Files

Rename your two XML files (`xmldata1.xml` and `xmldata2.xml`) in Assignment 1 according to the genre of the product. For example, if `xmldata1.xml` hosts products of the adventure genre, rename it to `adventure.xml`.

4.2 PHP Template for Reading XML Data

You will be writing a separate PHP template named `xmltemplate.php` which is stored in the *fevergamez/theme* directory. This template file will dynamically generate HTML pages (via query elements) from your two XML files (renamed `xmldata1.xml` and `xmldata2.xml`). The HTML pages that are generated should fit in with the style of the rest of your website (i.e., similar formatting, colors, etc).

Note that all of your XML files have the same tags and DTD. Thus, you only need to make 1 PHP template (`xmltemplate.php`) which allows the creation of a HTML page containing:

- A level 1 heading which mentions the genre.
- List of all the products in the genre, including information in your XML files such as:
 - Name
 - Description
 - Pricing
 - Search tags

- Reviews
- Any picture(s) of the product
- URL of a webpage related to the product.

Note that you are allowed to modify your XML files to cater for changes you need. *[Hint: refer to Lab 7 Task 2.]*

4.3 PHP File to Access Genre Pages

Create a PHP file named `genre.php` which is stored in the *fevergamez* directory. This is the "landing" page that will direct visitors to the correct genre pages by reading the name of the XML file from the GET parameters of the URL. As an example, to access the adventure genre HTML page, which is generated from *adventure.xml*, the URL may be `http://fevergamez.com/genre.php?genre=adventure`. In this case, the `genre.php` file is retrieving the query string 'adventure' from the URL. *[Hint: refer to Lab 7 Task 2.]*

Next, you need to include and update the navigation links on your website to enable access to these genre pages. Carefully note that all genre pages link to the same `genre.php` file, with subtle differences governed by the query string `?genre=` at the end of the URL.

Finally, in the genre pages, you should add JavaScript which allows you to store the name of the genre page viewed, along with the date and time it was viewed, in a cookie on the user's browser. Also, in the Homepage (`index.php`), add JavaScript which reminds a returning user of the product genre they were last viewing, along with the date and time they viewed it, by reading the stored cookie. *[Hint: refer to Lab 8 Task 3.]*

5. Data Collection Page

Create a Data Collection page named `collection.php`, where users can list and provide details about their second hand products. This will be done as a form, which sends the information as a POST request to a script `submit.php` located in the *fevergamez* directory when they click the "Submit" button. The following things need to be included on the page.

1. An appropriate level 1 heading.
2. Information about the person submitting:
 - Their first name.
 - Their last name.
 - Their email address.
 - A binary choice question on whether they have submitted a product before. Apply a default response here.
3. Information about the product - Include *all* information in your XML template, with the following additional requirements:
 - Required tags and attributes (elements with no modifiers or the '+' modifier, and attributes with #REQUIRED) should be required fields on your form before it can be submitted. These fields should be visually marked to the user with an asterisk '*'.
 - A dropdown box for the user to indicate the genre of the product. The dropdown box should list all existing genre.
 - A checkbox to indicate whether the user has read the terms and conditions. This checkbox should be required before the form can be submitted.
 - There should be buttons to add search tags for each product as per the XML. Each individual item should also have a button to delete it (there is a minimum of one, so the delete button of the last remaining item should be removed). Use JavaScript to implement the behaviour of these buttons. *[Hint: refer to Lab 8 Task 4.]*
 - If any reviews are entered, each review must contain the phrase "x out of 5 stars" (where x must

be a number from 0 to 5 allowing a single decimal place, i.e, 0, 1.1, 3, and 4.5 are all valid). Use JavaScript to check this when the submit button is clicked. The form should not submit if this phrase is not present in **each** review, and an error message should be displayed to the user. Remember, there could be zero reviews. *[Hint: refer to Lab 8 Task 2 and you may use GenAI tools to generate the regex string for check only.]*

- Buttons to submit and clear the form should be present.

Each field should use an appropriate input type that is intuitive to the user and minimizes the chance of an invalid input being entered.

Milestone 2

As you are progressing through [section 4](#) and [section 5](#), approach your lab instructor to check your second milestone during the labs between Week 10 and Week 12. The second milestone aims to check that you have started working on pulling data from your XML files, and you have a reasonable grasp of the basics of JavaScript.

The following will be checked:

- The PHP template file `xmltemplate.php` that pulls the XML data. This should correctly open the appropriate file and retrieve the data.
- Both of your genre pages. These pages are not required to be neat at this point, but you should be able to display all of the data from each file, including links and images, each in their appropriate elements (e.g., `<a>` and `<img>`).
- A working version of one of the yellow-highlighted JavaScript tasks in section 4 and 5. You only need to show one, but it must be fully functional.

Fulfilling the criteria listed above will grant you 1% of your Assessment 3 score.

6. Responsive Web Design

The site staff want the website to be well-designed, and easily accessible for both humans and computers (especially search engines). To that end, the website must look visually appealing across a wide range of devices, and use a consistent style independent of which web browser is used.

All webpages should use a responsive layout, that grows and shrinks when the screen size changes. The website should also use a different layout for small screens (and specifically mobile devices) and you must achieve this by using media queries in your `style.css`. You should set the *viewport* meta tag in your template files to indicate that your webpages are mobile-friendly. *[Hint: refer to Lecture 9.]*

In addition, you must optimise your website. This involves:

- Optimising images such that overly large files are not unnecessarily loaded for small display sizes. All images should be in a compressed, efficient format such as webp or avif. You can use [an image compression service](#) to optimise your images. All images must include a descriptive *alt* attribute that describes the image for screen readers.
- Minimizing content layout shift by setting explicit widths and heights for images.

7. [Bonus Task] Form Submission Script

You will write a PHP script `submit.php` to receive and process the data collection form (`collection.php`) for new products. Data from the form will be written to existing XML files according to the following specifications:

1. Once the script has confirmed that the required genre file exists, the XML file should be opened, and the new product data should be added. The resulting XML file **MUST** be valid (that is, it must validate).
2. At the end of a valid submission, the user should be redirected to the appropriate genre page showing their new product listed. A form submission success message should be displayed on this page to the user.

Successfully completing this task will earn you a 5 mark bonus.

8. Coding Style

All of your work (HTML, XML, CSS, PHP, and Javascript) should be validated and valid (meaning no errors). Use the following for validation:

- HTML: <https://validator.w3.org>
- XML: <https://www.xmlvalidation.com/>
- CSS: <http://jigsaw.w3.org/css-validator/>
- PHP: <https://phpcodechecker.com/>
- JavaScript: <https://beautifytools.com/javascript-validator.php>

Your HTML, XML, CSS, PHP, and JavaScript codes must be neat, clear, and indented. For all code files, you must include comments at the top of the file, stating the name of the file, the name of the author and the date the file was created. You must also at least include comments in your code files which addresses the marking criteria.

You must also include a README file named `README.txt` which includes all of your references.

9. Submission

Submission will be done via [Canvas](#). To submit your files, place the *fevergamez* directory into a `.zip` archive. The `.zip` archive should be named `A3_cXXXXXXX.zip` where `XXXXXXX` is replaced by **your** student id. Make sure your *fevergamez* directory contains the following:

- Homepage (`index.php`)
- About Us page (`about.php`)
- Contact page (`contact.php`)
- Genre 'landing' page (`genre.php`)
- Data Collection page (`collection.php`)
- XML template with DTD (`xmltemplate.xml` stored in the *fevergamez/data/xml* directory)
- XML data file 1 with DTD (renamed `xmldata1.xml` stored in the *fevergamez/data/xml* directory)
- XML data file 2 with DTD (renamed `xmldata2.xml` stored in the *fevergamez/data/xml* directory)
- CSS style file (`style.css` stored in the *fevergamez/theme* directory)
- PHP template file for HTML file creation (`template.php` stored in the *fevergamez/theme* directory)
- PHP template file for reading XML data (`xmltemplate.php` stored in the *fevergamez/theme* directory)
- Form submission script (`submit.php`)
- All images used for the website (stored in the *fevergamez/data/images* directory)
- README file (`README.txt`)

Note that failure to follow submission instructions will incur a penalty. Assignments submitted after the deadline will also be penalized by 10% per day late (including weekend). For example, an assignment worth 20% marked at 78% may be graded as:

- On time: final mark = $0.78 * 1.00 * 20 = 15.6$
- 1 day late: final mark = $0.78 * .9 * 20 = 14.04$
- 3 days late: final mark = $0.78 * .7 * 20 = 10.92$
- 5 days late: final mark = $0.78 * .5 * 20 = 7.8$

Extensions will only be granted with a valid and approved Adverse Circumstances application for the assignment. Adverse Circumstances application should also be submitted at least a day before the due.

Warning - Code Generation Tools

You are **NOT** allowed to use tools which generate HTML, XML, CSS, PHP, or JavaScript code, either from a visual representation or otherwise. This includes DreamWeaver, Microsoft Frontpage, GenAI tools (e.g., ChatGPT), saving to HTML from MS Word, etc. You must write all code yourself in a **plain-text** editor, such as Notepad++ or VS Code (with Live Server extension only). Any assignments that use code generated from a tool will be given **ZERO**.

Warning - Plagiarism

Plagiarism is defined as presenting the work of others' as if it was your own. This includes the copying of content from other people's web pages (and other sources) without "appropriate referencing" (indicating the parts of your assignment which have been copied and where they were copied from).

If you "quote" or use other people's work (or images) in your assignments, then you **must** reference them appropriately (using a README.txt file). Failure to do so will cause you to lose marks for the assignment. An assignment that consists largely of work copied from elsewhere will receive very low (zero) marks, and may be reported to the school for suspicion of plagiarism.

Warning - GenAI Usage

The use of GenAI tools (such as ChatGPT) is forbidden for the writing of any code. However, generative AI may be used for purely creative aspects of the website (such as coming up with descriptions, welcome text, and images) or wherever explicitly stated. AI generated content should also be referenced appropriately (using a README.txt file). AI detection tools will check each file, and any content with a sufficiently high detection score will be reported to the School for suspicion of plagiarism.

You can find more information on the University's Policy on Student Academic Integrity in the Course Outline and at [Policy Library](#).

10. Marking Guide

The following serves as a rough guide to how many marks are allocated to each section of the assignment. Please refer to the rubrics on Canvas for a complete breakdown.

Total Marks: 92

- [General Requirements](#): 12
- [Web Server Setup and Website Import](#): 6.5
- [Genre Pages](#): 13.5

- [Data Collection Page](#): 22
- [Responsive Web Design](#): 6
- [Coding Style](#): 32
- [\[Bonus Task\] Form Submission Script](#): [5]
- You may be penalised for poor aesthetic appearance, untidiness, spelling and grammar problems, and not correctly following instructions: [-10]